

Results: Of 216 total patients, 69 initially underwent subtotal resection (Simpson Grade IV or V). Thirteen of these 69 patients received postoperative radiotherapy and had a 5-year RFS of 77% compared to 37% for those who did not (p value = 0.031). The resulting hazard ratio was 0.42 (95% CI, 0.19-0.95) for recurrence or death with post-operative radiation after subtotal resection. In total, 78 of 216 patients had at least one recurrence. Among those patients with recurrent disease, 46 received radiation: 33 received postoperative radiation, 9 received radiation alone, and 4 received both radiation alone and postoperative radiation over the course of multiple recurrences. The median initial dose of postoperative radiation was 54 Gy (range, 9-60 Gy) and for radiation alone was 50 Gy (range, 13-54 Gy). Of the 46 patients who received any radiation for recurrent disease, 11 did not recur again after radiation; of the 35 who did recur, the median average interval to recurrence after radiation was 2.4 years compared to 3.7 years before radiation (p = 0.075).

Conclusions: Patients who receive postoperative radiation after initial subtotal resection have a lower risk of recurrence or death. Among patients who receive radiation for recurrent disease, there appears to be two populations: patients who never recur again, and patients who recur at least as fast as they did prior to radiation. Known pathologic characteristics will be explored in combination with genetic studies to identify molecular features that distinguish these populations.

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2158 Outcomes after ^{125}I Plaque Brachytherapy for Ocular Melanoma: The Yale Cancer Center Experience

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Purpose/Objectives: To determine the outcome of ^{125}I plaque brachytherapy for primary ocular melanoma tumors at the Yale Cancer Center. To compare the outcomes for medium vs. large tumor size and the impact of transpupillary thermotherapy (TTT), as an adjunct, on outcome, and toxicity.

Materials/Methods: We retrospectively reviewed the characteristics and outcomes of 38 patients who received primary ^{125}I plaque brachytherapy for ocular melanoma at the Yale Cancer Center (YCC) between February 1993 and June 2005; follow-up data were available for 27 patients who are included in this analysis. Patients were divided into two categories, those with medium (M) tumors measuring 2.5-10 mm in apical height or a median basal tumor diameter (MBTD) $\leq 16\text{mm}$ and those with large tumors (L) measuring $> 10\text{ mm}$ in height or an MBTD > 16 . Patient charts were reviewed for pre- and post-procedure visual acuity, local recurrence, subsequent enucleation, toxicity, and survival. Survival data was obtained from patient charts as well as the Social Security Death Index Database.

Results: Median age at treatment was 54 years (range, 30-89). Median follow-up time for the cohort was 3.7 years. 63% had ≥ 2 -year follow-up and 33% had ≥ 5 -year follow-up. The median tumor dimensions were $17 \times 13 \times 7.5\text{ mm}$. Eleven (40.7%) medium-sized tumors and 16 (59.3%) large tumors were implanted. Patients were treated with a median dose of 97 Gy to the tumor apex at a median dose rate of 82.8 cGy/hour. A TTT was performed in 8 patients, 2 in the M cohort and 6 in the L cohort. Enucleation was performed in 5 patients, all in patients with large tumors. Four of 5 (80%) were successfully salvaged and alive at time of analysis. Five-year survival rates were 80% for the 20 patients treated ≥ 5 -years-ago. Median survival was 6 years, 9.8 years in the M group, 3.5 years for L group. There were 3 local recurrences, all which occurred in the L cohort. Two were successfully salvaged (1 with TTT, the other with enucleation and TTT) and were alive at the time of analysis. Median visual acuity was 20/50 pre-treatment and median visual acuity post-implant was 20/400. The most common side effects for all patients were cataract formation (9/27), vitreous hemorrhage (5/27), and retinopathy (7/27). Four patients underwent elective enucleation without evidence of local recurrence.

Conclusions: ^{125}I plaque brachytherapy for ocular melanoma at the YCC showed results comparable to previously published data in terms of visual acuity, local control, survival, and toxicity. Results for patients with medium-sized uveal melanomas were particularly excellent. While plaque brachytherapy for large-sized uveal melanoma is associated with greater local recurrence, it is a feasible alternative to enucleation.

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2159 Histone Deacetylase Inhibitory Activity of Valproic Acid in Glioblastoma

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Purpose/Objective(s): To evaluate the efficacy and toxicity of histone deacetylase inhibitor (HDACi), valproic acid (VPA) and radiotherapy in newly diagnosed glioblastoma multiforme.

Materials/Methods: Between February 2002 and November 2006, 73 patients (median age, 53 years; range, 11-78 years) with histologically confirmed glioblastoma multiforme were treated with surgery followed by radiotherapy with or without chemotherapy. Intravenous or oral VPA was administered to 66 patients. Surgical extents were gross total resection in 19 patients (26%), subtotal resection in 35 (48%), and biopsy only in 19 (26%). The median radiation dose was 59.4 Gy, ranging from 52.2 to 61.2 Gy.

Results: Four (6%) had a complete response (CR), 19 (26%) partial response (PR), 24 (33%) stable disease (SD), and 22 (30%) progressive disease (PD) after radiotherapy with or without VPA. Median time to progression was 10 months and median survival was 25 months. Median overall survival was 25 months in VPA-treated patient group and 8 months in patient group treated without VPA (p = 0.02). Favorable prognostic factors by univariate analyses on progression free survival were no neurologic symptom (p = 0.01), ECOG performance status 0 or 1 (p = 0.01), existence of RT response (p < 0.01). Favorable